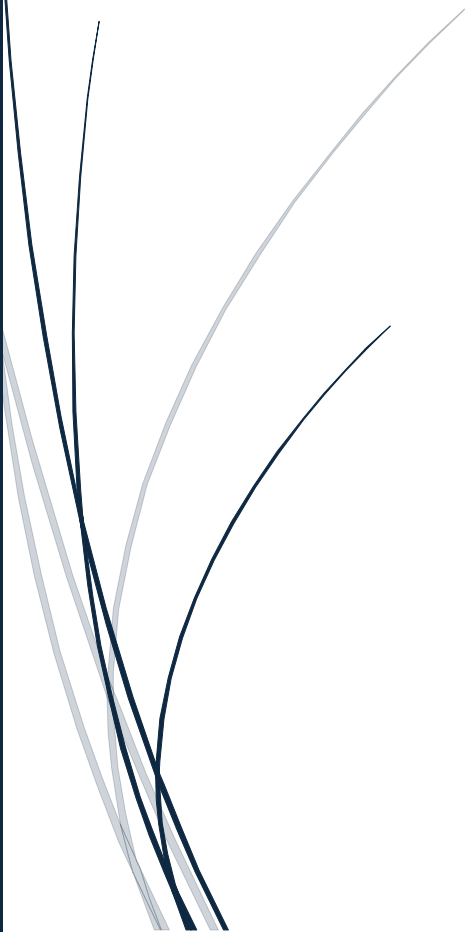


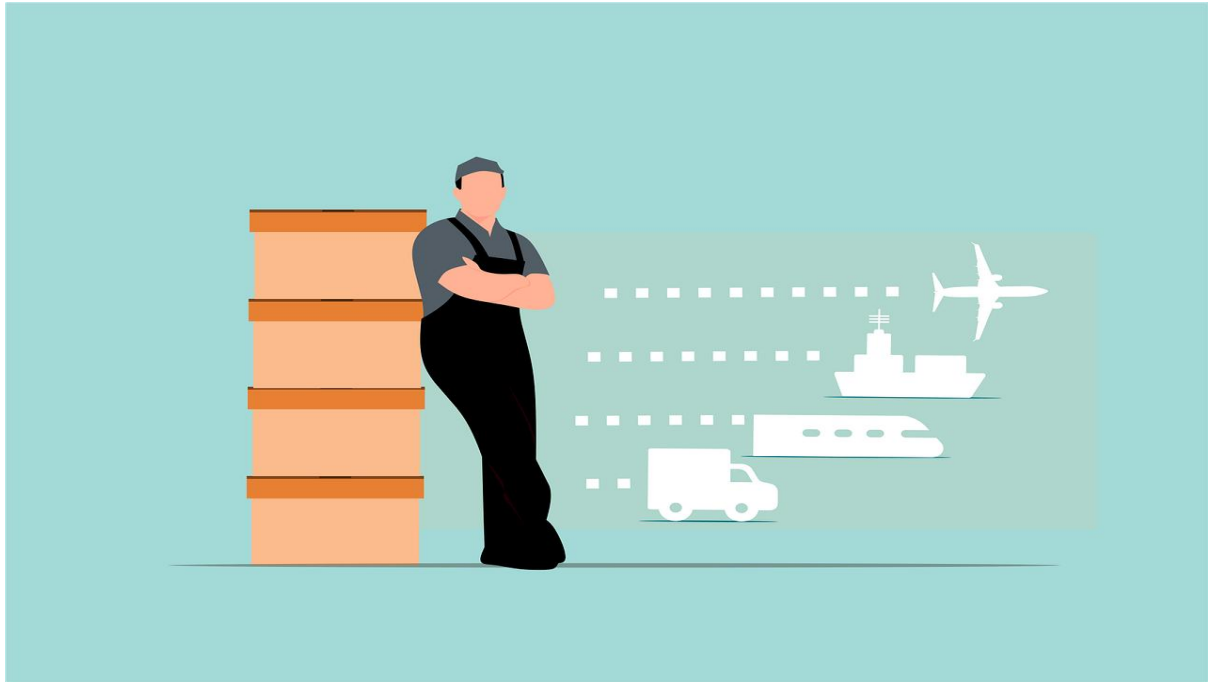


Inventory Optimization for Retail

MySQL Data Analysis Project Report



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1. Introduction

Efficient inventory and pricing management are essential for retail success.

This project analyses retail data to identify **demand patterns, pricing effectiveness, and inventory efficiency.**

The goal is to:

- Optimize product availability
- Reduce stockouts and overstocking
- Analyses pricing strategies and their effect on sales
- Support data-driven decision-making using MySQL

This dataset is designed to support the development of a **multi-agent AI system** for optimizing inventory management in the retail industry. By integrating data from stores, warehouses, suppliers, and customers, this dataset enables AI-driven demand forecasting, inventory tracking, and pricing optimization to **reduce stockouts, minimize holding costs, and improve supply chain efficiency.**

Dataset Composition

- **Sales Data:** Historical transaction records, product sales trends, and seasonal demand fluctuations.

- **Inventory Levels:** Real-time stock availability at stores and warehouses, including restocking frequency.
 - **Supplier & Logistics Data:** Delivery schedules, lead times, and supplier performance metrics.
 - **Pricing & Promotions:** Historical price changes, discount strategies, and customer responses.
- This dataset empowers AI models to **predict demand, automate inventory tracking, optimize pricing strategies, and enhance overall retail efficiency**, ensuring the right products are available at the right time while minimizing operational costs.

2. Data Description:

1. Demand for Casting (Sales & Demand Data) :

Used for sales trend analysis, demand forecasting, and promotion impact assessment.

Column Name	Description	Type	Example
Product ID	Unique identifier for each product in the catalog. Used to link across all datasets.	Integer / String	P001
Date	Date of sale or record entry, useful for time-series and trend analysis.	Date	2025-05-10
Store ID	Unique identifier for each retail store or location.	Integer / String	S012
Sales Quantity	Number of product units sold on that date at that store.	Integer	320
Price	Selling price per unit of the product at the time of sale.	Decimal	49.99
Promotions	Indicates if any promotional offers or discounts were active (e.g., “Yes/No” or “Type”).	String	Festival Offer, No
Seasonality Factors	Seasonal elements that affect sales (like festive seasons, summer demand, etc.).	String / Category	Holiday, Winter

External Factors	Outside influences like weather, economy, or events affecting demand.	String / Category	Rainy, Economic Slowdown
Demand	Estimated or observed customer demand for the product (may differ from sales if stockouts occur).	Decimal	500
Trend	Direction of demand or sales change over time (e.g., increasing, stable, declining).	String	Upward
Customer Segments	Category of customers targeted or purchasing (e.g., youth, premium, budget shoppers).	String	Premium, Family

2. Inventory Monitoring (Stock & Supply Chain Data) :

Used for inventory management, supplier performance evaluation, and stock optimization.

Column Name	Description	Type	Example
Product ID	Unique product identifier (linked with other tables).	Integer / String	P001
Store ID	Unique store/location identifier.	Integer / String	S012
Stock Levels	Current available inventory units at the store or warehouse.	Integer	450
Supplier Lead Time (days)	Average number of days suppliers take to deliver new stock after ordering.	Integer	7
Stockout Frequency	How often a product runs out of stock (e.g., per month).	Integer	3
Reorder Point	Minimum stock level that triggers reordering to avoid stockouts.	Integer	200
Expiry Date	The date when the product expires (for perishable goods).	Date	2025-12-31
Warehouse Capacity	Total number of units the warehouse/store can hold for that product.	Integer	1000

Order Fulfillment Time (days)	Time taken to process and complete customer orders after they are placed.	Integer	2
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3. Pricing Optimization (Pricing & Market Data) :

Used for pricing strategy optimization, competitive benchmarking, and profitability analysis.

Column Name	Description	Type	Example
Product ID	Product identifier (links to the same product in other datasets).	Integer / String	P001
Store ID	Store identifier.	Integer / String	S012
Price	Actual selling price per unit in the current pricing strategy.	Decimal	49.99
Competitor Prices	Price of the same or similar product sold by competing stores.	Decimal	52.50
Discounts	Discount offered (absolute or percentage).	Decimal	10% or 5.00
Sales Volume	Total quantity sold in a specific period (daily, weekly, etc.).	Integer	1200
Customer Reviews	Average customer rating (e.g., from 1–5 scale).	Decimal	4.3
Return Rate (%)	Percentage of sold items returned by customers.	Decimal	2.5
Storage Cost	Cost of storing unsold inventory (per unit or total).	Decimal	1.25
Elasticity Index	Measures how sensitive demand is to price changes (elasticity > 1 = highly price-sensitive).	Decimal	1.8

3. Database Schema Design (MySQL):

Relationships:

- All three tables are linked by product_id and store_id.
- This enables multi-table joins for comprehensive analysis.

demand_casting (product_id, store_id)

↓

inventory_monitoring (product_id, store_id)

↓

pricing_optimization (product_id, store_id)

4. Data Loading and Cleaning - Steps

1. Create Database
2. Import CSV Files
3. Check Data Quality

5. Data Integration (Joins)

Create a view combining all datasets

6. Data Analysis Using MySQL Queries

1. Top Selling Products
2. Products with High Demand but Low Stock
3. Promotion Effectiveness
4. Price Elasticity Analysis
5. Supplier Lead Time vs Fulfilment Efficiency
6. Key Performance Indicators (KPIs)

7. Key Performance Indicators (KPIs)

KPI	SQL Formula	Insight
Total Sales	SUM(sales_quantity)	Total quantity sold
Stock Efficiency	total_demand / stock_levels	Balance between demand and inventory
Promotion ROI	avg_sales_promo / avg_sales_no_promo	Effectiveness of promotions
Average Lead Time	AVG(supplier_lead_time)	Supplier performance
Price Elasticity	AVG(elasticity_index)	Sensitivity of sales to price changes

8. Insights & Findings

- Demand-Inventory Mismatch:**
Certain products show **high demand but insufficient stock**, indicating a need for improved forecasting.
- Promotion Impact:**
Promotions significantly increase sales quantities, particularly during **seasonal events**.
- Pricing Strategy:**
Products with an **elasticity index > 1** show strong price sensitivity — lowering prices here boosts sales.
- Inventory Optimization:**
High **lead times and frequent stockouts** suggest supplier or logistics delays impacting availability.
- Customer Behaviour:**
Stores with higher customer ratings also exhibit **lower return rates**, implying quality or service consistency.

9. Conclusion

By leveraging MySQL queries and multi-table joins, this project identifies:

- Which products need better inventory control
- How pricing and promotions affect demand
- How to align stock with expected sales

The analysis provides actionable insights to:

- Optimize restocking and supplier scheduling
- Design effective pricing strategies
- Improve overall retail performance